DATA QUALITY & INTEGRITY AUDIT REPORT

Project: E-Commerce Transaction Data Cleaning | **Client:** Portfolio Delivery

1. Executive Summary

Comprehensive data cleaning and validation conducted on the **Online Retail Dataset** (541,909 raw records). Primary anomalies included duplicate logs, non-commercial warehouse write-offs, invalid zero/negative pricing, and unassigned customer profiles. The remediation pipeline retained **534,129 valid commercial records (98.56% retention)** with **0 missing values**, reconciling a net commercial revenue of **£9.75M** while preserving accounting traceability for £894K in returns.

2. Pre-Cleaning vs. Post-Cleaning Audit Benchmark

Metric / Dimension	Raw Dataset	Cleaned Dataset	Remediation Action & Justification
Total Records	541,909	534,129	98.56% Valid transactions preserved
Total Columns	8	11	+3 Analytical columns engineered
Duplicate Records	5,268	0	100% Purged (5 had zero unit price)
Missing Customer IDs	135,080	0	Retained & assigned unique Guest IDs
Missing Descriptions	1,454	0	Resolved via non-positive price filtering
Warehouse Loss / Damage	1,336	0	Verified zero price & null client; purged
Non-Positive Prices (<= 0)	2,517	0	2,512 purged post-dedup (incl. -£11k debt)
Net Realized Revenue	Distorted	£9,748,131.07	Reconciled (Gross £10.64M - £894K Returns)
Invoice Date Field	String / Object	datetime64[ns]	Converted & validated to UTC timestamps

3. Core Cleaning Strategy & Business Justifications

- **Deduplication:** 5,268 identical rows removed to prevent double-counting orders and skewing inventory.
- **Audit of Negative Quantities:** Of 10,587 negative records, 9,251 were legitimate customer cancellations ('C' prefix) and preserved. The remaining 1,336 had 0.0 unit price, missing CustomerIDs, and annotations like 'damaged' or 'check'—confirmed internal warehouse loss, not sales.
- **Zero & Negative Price Treatment:** 2,517 non-positive price rows in raw data (5 duplicate, 2,512 filtered post-deduplication, including two £-11,062 bad debt adjustments) removed to maintain sales accuracy.
- **Guest Checkout Preservation (£1.47M):** 132,565 guest sales (15.08% of revenue) preserved by imputing structured identifiers (Guest_<InvoiceNo>) and flagged via Is_Guest.
- **Text Normalization:** Over 112,000 product descriptions sanitized from irregular padding spaces.

4. Engineered Columns & Downstream Capabilities

- **TotalPrice:** Calculated as Quantity * UnitPrice for immediate sales aggregations.
- **Is_Cancelled:** Boolean indicator distinguishing refunds from gross sales.
- **Is_Guest:** Boolean indicator facilitating registered-user filtering for customer segmentation (RFM).